Course : AI Assistant Coding

Instructor : Sallauddin Mohammed

Title : Car Rental System

Team Details : D. Sai Siri Reddy (2303A51236)

Introduction :

The Car Rental System is a full-stack web application that allows users to:

- Browse available cars
- Book cars with drivers
- View booking history

The system also includes an **Admin Panel** to manage:

- Cars
- Drivers
- Bookings

Objective :

Build a full-stack application using MERN stack

- Provide seamless car booking experience
- Implement authentication using JWT
- Manage resources (cars, drivers) efficiently
- Apply AI-assisted development for faster coding

Stakeholders :

- **Users:** Book cars
- **Admin:** Manage cars, drivers, bookings
- **Developers:** Maintain system
- **Business Owner:** Revenue from bookings

Technologies Used :

Frontend:

- React.js
- CSS (inline styling)

Backend:

- Node.js
- Express.js

Database:

- MongoDB

Tools:

- GitHub
- VS Code

Authentication:

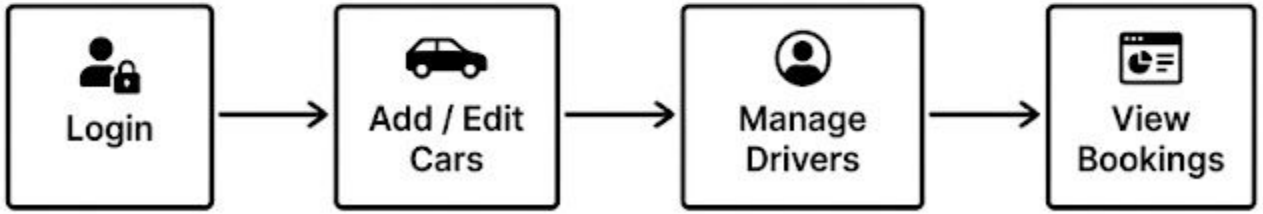
- JWT (JSON Web Token)

AI Prompts Used :

- Create a React car booking page with form and API integration”
- “Fix 401 unauthorized JWT error in MERN app”
- “Improve UI for car cards with premium styling”
- “Create Admin dashboard with CRUD operations”
- “Filter available drivers in booking system”

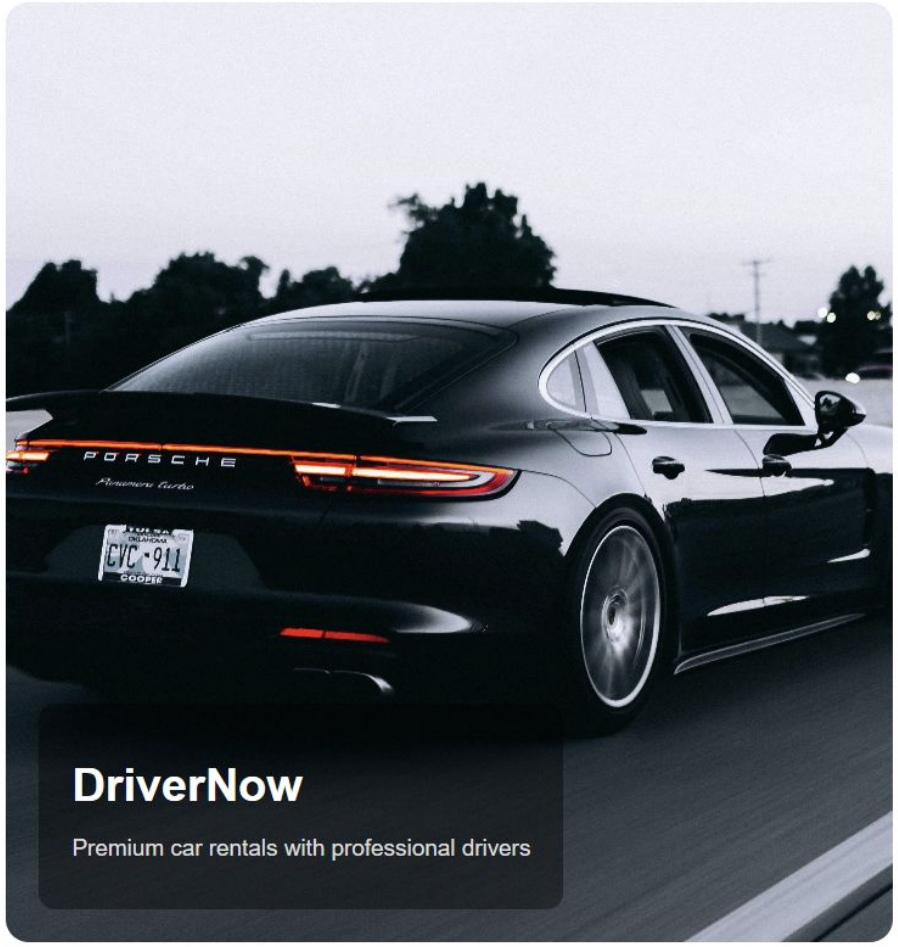
USER WORKFLOW

ADMIN WORKFLOW



DriverNow

Home Cars Login Get Started



Create Account

Start your journey in seconds

Full Name

Email

Password

Create Account

Already have an account? **Login**

DriverNow

Home Cars **Admin**

Logout

Admin

CARS

DRIVERS

BOOKINGS

Search...

Cars

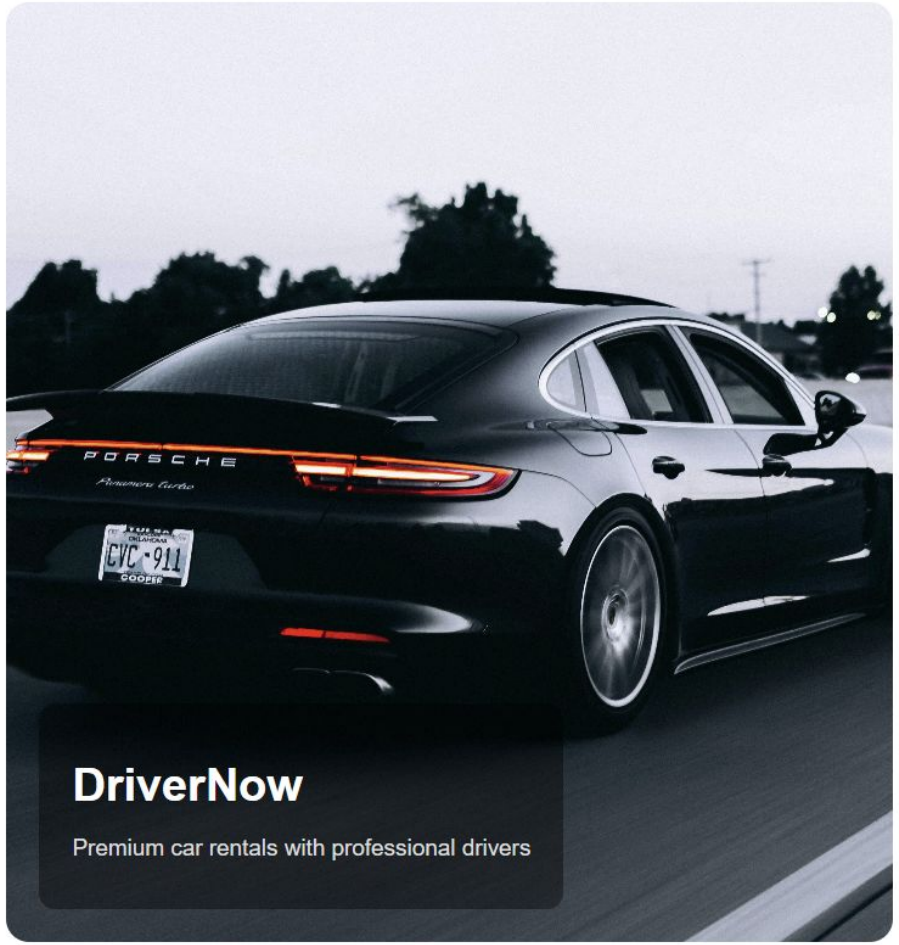
Name | Price/km | Base Rent | Category | Add Car

- Mercedes-Benz C-Class (luxury) - ₹450/km — Delete
- BMW 3 Series (luxury) - ₹450/km — Delete
- Toyota Fortuner (luxury) - ₹300/km — Delete
- Mahindra XUV700 (suv) - ₹140/km — Delete
- Kia Seltos (suv) - ₹130/km — Delete
- Hyundai Creta (suv) - ₹130/km — Delete
- Maruti Swift Dzire (sedan) - ₹110/km — Delete
- Hyundai Aura (sedan) - ₹100/km — Delete

DriverNow

Home Cars Admin

Logout



Welcome to DriverNow

DriverNow is a modern car rental platform designed to provide seamless and affordable travel experiences. Whether you need a quick city ride or a long-distance journey, we offer a wide range of vehicles with professional drivers.

✓ Wide range of cars (Economy → Luxury)

✓ Transparent pricing

✓ Verified drivers

✓ Easy booking system

Explore Cars

DriverNow

Home Cars Dashboard

Logout

Explore Cars

Find the perfect ride for your journey

Search cars...

ALL ECONOMY SEDAN SUV LUXURY

ECONOMY

Renault Kwid Available
ECONOMY
4 Seats Petrol Manual
Base ₹900
₹90/km
Book Now

Maruti Alto 800 Available
ECONOMY
4 Seats Petrol Manual
Base ₹800
₹80/km
Book Now

Datsun Redi-GO Unavailable
ECONOMY
4 Seats Petrol Manual
Base ₹1000
₹60/km
Unavailable

Maruti Alto K10 Unavailable
ECONOMY
4 Seats Petrol Manual
Base ₹700
₹70/km
Unavailable

Renault Kwid 2 Available
ECONOMY

DriverNow

Home Cars Dashboard

Logout

Book Now

Book Now

LUXURY

Mercedes-Benz C-Class Available
LUXURY
4 Seats Petrol Manual
Base ₹5000
₹450/km
Book Now

BMW 3 Series Available
LUXURY
4 Seats Petrol Manual
Base ₹5000
₹450/km
Book Now

Toyota Fortuner Available
LUXURY
4 Seats Petrol Manual
Base ₹4000
₹300/km
Book Now

Skoda Superb Available
LUXURY
4 Seats Petrol Manual
Base ₹4000
₹280/km
Book Now

Mercedes-Benz C-Class Available
LUXURY
4 Seats Petrol Manual
Base ₹5200
₹360/km
Book Now

Audi A4 Unavailable
LUXURY
4 Seats Petrol Manual
Base ₹4800
₹320/km
Unavailable

DriverNow

Home Cars **Dashboard** Logout

My Dashboard

5

Total Bookings

Skoda Slavia ₹2250

WGL → HYD

Driver: Suresh Babu

Confirmed View Details →

Toyota Hyryder ₹2650

HYD → WGL

Driver: Sneha Patel

Confirmed View Details →

Datsun Redi-GO ₹1300

HYD → WGL

Driver: Samanvith

Confirmed View Details →

Audi A4 ₹12800

Warangal → Karimnagar

Driver: Amit Kumar

Confirmed View Details →

Maruti Alto K10 ₹1750

Hanamkonda → Hyderabad

Driver: Priya Reddy

Confirmed View Details →

Audi A4

Route:
Warangal → Karimnagar

Driver: Amit Kumar

Total: ₹12800

Payment

Card Number

MM/YY

CVV

Pay ₹12800

Proof Of AI Prompts :

File Edit Selection View Go Run ... car-rental-system

EXPLORER

CAR-RENTAL-SY...
- .github\workflows
  - ci.yml
- client
  - node_modules
  - public
  - src
    - api
    - assets
    - components
      - BookingForm.jsx
      - CarCard.jsx
      - DriverSelect.jsx
      - Navbar.jsx
      - ProtectedRoute.jsx
    - pages
    - services
    - App.css
    - App.jsx
    - index.css
    - main.jsx
  - .gitignore
  - eslint.config.js
  - index.html
  - package-lock.json
  - package.json
  - README.md

OUTLINE
TIMELINE
MYSQL

ulatePrice.js | calculateDistance.js | bookingController.js | Dashboard.jsx | BookingForm.jsx | api.js ...\

client > src > components > BookingForm.jsx > default

You, 5 days ago | 1 author (You)

```
// Create a React component for a car booking form that allows users to enter trip details
// such as pickup location, drop location, start date, and end date. Use useState to manage
// form inputs and selected driver ID, and use useEffect to fetch available drivers from the API
// when the component loads. Integrate a DriverSelect component to display and choose a driver.
import { useEffect, useState } from "react";
import axios from "../api/api";
import DriverSelect from "./DriverSelect";
import { useNavigate } from "react-router-dom";

const BookingForm = ({ carId }) => {
  const navigate = useNavigate();

  const [drivers, setDrivers] = useState([]);
  const [driverId, setDriverId] = useState("");
  const [car, setCar] = useState(null); // ✅ NEW

  const [form, setForm] = useState({
    pickupLocation: "",
    dropLocation: "",
    startDate: "",
    endDate: ""
  });

  const [distance, setDistance] = useState(0);
  const [price, setPrice] = useState(0);
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | GITLENS | POSTMAN CONSOLE | node - client

```
PS C:\Users\dasar\car-rental-system\client> npm run dev
  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

```
PS C:\Users\dasar\car-rental-system\server> npm run dev
MongoDB Connected
Server running on port 5000
```

File Edit Selection View Go Run car-rental-system

EXPLORER

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    - App.css
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  - package.json
  - README.md
- OUTLINE
- TIMELINE
- MYSQL

CarCard.jsx | Navbar.jsx | DriverSelect.jsx | ProtectedRoute.jsx | index.css | authMiddleware.js | User.js

client > src > components > CarCard.jsx > default

```jsx
// Create a reusable React component to display car details in a card format.
// Accept a car object as a prop and safely handle cases where the car data may be undefined.
// Show the car name, availability status, base rent, and price per kilometer with clean UI styling.
import { useNavigate } from "react-router-dom";

const CarCard = ({ car }) => {
  const navigate = useNavigate();

  if (!car) return null;

  const handleBooking = () => {
    if (car._id) {
      navigate(`/booking/${car._id}`);
    }
  };

  return (
    <div style={styles.card}>

      {/* HEADER */}
      <div style={styles.top}>
        <div>
          <h3 style={styles.name}>{car?.name || "Unnamed Car"}</h3>
          <p style={styles.category}>
            {car?.category?.toUpperCase() || "ECONOMY"}
          </p>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS POSTMAN CONSOLE

node - client

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File Edit Selection View Go Run car-rental-system

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- MYSQL

CarCard.jsx | Navbar.jsx | DriverSelect.jsx 1 | ProtectedRoute.jsx | index.css | authMiddleware.j

client > src > components > DriverSelect.jsx > default

You, 5 days ago | 1 author (You)

```
// Create a reusable React component for selecting a driver from a dropdown list.
// Accept a list of drivers and a setter function as props to update the selected driver ID.
// Render a select input with a default "Select Driver" option.
// Create a reusable React component for selecting a driver from a dropdown list.
// Accept a list of drivers and a setter function as props to update the selected driver ID.
// Render a select input with a default "Select Driver" option.
import { useState, useEffect } from "react";
import axios from "../api/api";

const DriverSelect = ({ setDriverId }) => {
  const [drivers, setDrivers] = useState([]);
  const [selected, setSelected] = useState("");

  useEffect(() => {
    fetchDrivers();
  }, []);

  const fetchDrivers = async () => {
    try {
      const res = await axios.get("/drivers");

      // ✅ only available drivers
      const available = res.data.filter((d) => d.available === true);

      setDrivers(available);
```

PROBLEMS 2 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | GITLENS | POSTMAN CONSOLE | node - client

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MongoDB Connected
Server running on port 5000
```

References

- React Documentation
- MongoDB Docs
- Node.js Docs
- Express.js Docs
- ChatGPT (OpenAI)
- Stack Overflow

Github Link

- https://github.com/saisiridasari/Car-Rental-System

THANK YOU!